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GS 95011-2:2018-12

Electrical and electronic components in motor vehicles

Assembly and connection technology in motor vehicle electronics

Production and testing

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Foreword

The German version is binding.

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Amendments

The following amendments have been made to GS 95011-2:2018-12:

- section 4 updated, is now section 3;
- sections 5, 5.1.1, 5.1.3, 5.4.4.3, 5.4.5, 5.7, 5.10.2, 5.10.7, 6.2, 9.1.3 updated;
- table 2 "Die attach in the power module" updated, is now Table 1;
- section 5.10.3 restructured;
- section 6.7 added;
- normative references updated;
- Group Standard editorially revised.

Previous editions

BMWN(S) 60122.0 T(P)2: 1989-01, 1994-01, 2001-10

GS 95011-2: 2001-10, 2005-12, 2009-05, 2012-02, 2013-06, 2018-12

1 Scope and purpose

This Group Standard defines the basic requirements for motor vehicle electronics and is to be applied in the case of new developments, new releases of existing assemblies due to changes and the use of new technologies. This Group Standard is type-independent and generally applies to all unfitted and fitted circuit carriers as well as electric, electronic and mechatronic units.

2 Normative references

This Group Standard incorporates provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. The respective latest edition of the publication is applicable.

DIN EN 60664-1	Insulation coordination for equipment within low-voltage supply systems; Part 1: Principles, requirements and tests
DIN EN 60664-1 SUPPL1	Insulation coordination for equipment within low-voltage supply systems; Part 2-1: Application guide; Explanation of the application of the IEC 60664 series, dimensioning examples and dielectric testing
DIN EN 60664-4	Insulation coordination for equipment within low-voltage supply systems; Part 4: Consideration of high-frequency voltage stress
DIN EN 61340-5-1	Electrostatics; Part 5-1: Protection of electronic devices from electrostatic phenomena; General requirements
DIN EN 61340-5-1 SUPPL1	Electrostatics; Part 5-2: Protection of electronic devices from electrostatic phenomena; User guide
DVS 2811	Test Procedures for Wire Bonded Joints
GS 93008-1	Substances of concern; Materials and components; Prohibited and declarable substances
GS 93008-2	Substances of concern; Process materials and auxiliary materials; Prohibited and declarable substances
GS 93008-4	Substances of concern; Materials and components; Constituents and emissions
GS 93024	Recycling of motor vehicles; Recycling-optimized vehicle design
GS 95009	Electrostatics; protective measures against electrostatic discharge (ESD)
GS 95017	Traceability of electrical and electronical components and units; Requirements for the supplier
GS 95011-1	Electrical and electronic components in motor vehicles; Assembly and connection technology in motor vehicle electronics; Technology and layout
GS 95024-3-1	Electrical and electronic components in motor vehicles; Environmental requirements and testings Technical test specifications
GS 95024-3-3	Electrical and electronic units in motor vehicles; Run-in of units
GS 95039	Electric and electronic units for motor vehicles; Hardware-Delta-Qualification-Matrix; Delta-qualification of units during series changes
GS 98000	Statistical capability studies
IPC-A-610	Acceptability of Electronic Assemblies

IPC-2221	Generic Standard on Printed Board Design
IPC-2222	Sectional Design Standard for Rigid Organic Printed Boards
IPC-2223	Sectional Design Standard for Flexible/Rigid-Flexible Printed Boards
IPC-7526	Stencil and Misprinted Board Cleaning Handbook
IPC-7711/21	Rework, Modification and Repair of Electronic Assemblies
IPC-7912	Calculation of DPMO and Manufacturing Indices for Printed Board Assemblies
IPC-9261	Calculation of In Process DPMO and Manufacturing Indices for Printed Board Assemblies
IPC-9704	Printed Circuit Assembly Strain Gage Test Guideline
JESD201	Environmental acceptance requirements for tin whisker; Susceptibility of tin and tin alloy surface
J-STD-001	Requirements for Soldered Electrical and Electronic Assemblies
UL 94	Tests for flammability of plastic materials for parts in devices and appliances
VDA Volume 19 T1	Quality Management in the Automotive Industry; Inspection of Technical Cleanliness; Particulate contamination of functionally-relevant automotive parts
VDA Volume 19 T2	Quality Management in the Automotive Industry; Technical cleanliness in assembly; Environment, logistics, personnel and assembly equipment
ZVEI	Guideline for Technical Cleanliness
1907/2006[1)]	Regulation (EC) No 1907/2006 of the European Parliament and of the council of 18 December 2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH), establishing a European Chemicals Agency, amending Directive 1999/45/EC and repealing Council Regulation (EEC) No 793/93 and Commission Regulation (EC) No 1488/94 as well as Council Directive 76/769/EEC and Commission Directives 91/155/EEC, 93/67/EEC, 93/105/EC and 2000/21/EC
2000/53/EC[)]	Directive 2000/53 of the European Parliament and of the Council dated 18 September 2000 on end-of-life vehicles

1) See www.umwelt-online.de / "umwelt-online Login"

3 Abbreviations and definitions

3.1 Abbreviations

3D MID Molded interconnect device
AOI Automated optical inspection)
AVT Assembly and connecting technology
AXI Automated X-ray inspection
BGA Ball grid array
DFN Dual flat no leads
DoE Design of experiments
DPMO Defects per million opportunities
ELV End-of-life vehicles (End-of-life vehicle directive 2000/53/EU)
EMC Electromagnetic compatibility
ESD Electrostatic discharge
FR4 Flame retardant, according to UL 94 V-0
HAL Hot air leveling
HV High voltage
HW-DQM Hardware delta qualification matrix
IC Integrated circuits
ICT In-circuit test
KPI Key performance indicator
LCD Liquid crystal display
LED Light-emitting diode
MLCC Multi-layer ceramic capacitor
MOST Media-oriented systems transport (bus)
PCB Printed circuit board: unfitted printed circuit board

PI	Polyamide
PEN	Polyethylene naphthalate
PET	Polyethylene terephthalate
PWB	Printed wiring board: unfitted printed circuit board
QFN	Quad flat no leads
QMT	Quality management parts
REACH	Registration, Evaluation, Authorisation and Restriction of Chemicals (Ordinance 1907/2006)
Reman	Remanufacturing
SMD	Surface mounted device
SOP	Start of production
SPC	Statistical process control
SPI	Solder paste inspection

3.2 Definitions

Level	Designation	AVT	EXAMPLE
1	Chip level	-	Silicon chip (inside module)
2	Component level	1st level	Component, circuit carrier
3	Assembly level	2nd level	Assembled circuit board
4	Unit level	3rd level	Complete control unit
5	Vehicle level	-	-

4 General

4.1 General requirements

All relevant legal regulations and laws shall be complied with for safety requirements, certification (especially exhaust emissions) and quality. In addition, the relevant BMW requirements apply.

In terms of ingredients and recyclability, the materials, procedure and process technology, parts, and systems shall meet all applicable legal requirements.

Each assembly containing electronic components shall be marked as such.

4.2 Basic requirements

Deviations from this Group Standard shall be agreed with the responsible specialist department and approved by the responsible QMT. They shall be documented in the requirements specification.

Deviations that only become evident in the course of development shall be documented in the technical drawings.

4.3 Requirements for materials and processes

Only the use of qualified and approved materials is permissible.

Assembly and production are carried out based on the current state of the art.

4.4 Material bans, environment, recycling and sustainability

The environmental, sustainability and social standards defined in the international BMW purchasing conditions (IPC) shall be complied with.

The material restrictions (specifications by the European Union, e.g. Directive 2000/53/EC (ELV) and Ordinance 1907/2006 (REACH)) and associated reporting requirements in accordance with GS 93008-1, GS 93008-2 and GS 93008-4 shall be observed throughout the entire product life cycle.

Chemicals and waste water shall be purified according to the state of the art.

The following requirements shall be observed during development of new circuit carriers/electrical and electronic units:

- reducing the variety of plastics;
- avoiding harmful, toxic substances;
- using halogen-free or halogen-reduced materials;
- minimizing weight of poorly recyclable parts;
- avoiding waste-intensive and energy-intensive production processes involving harmful substances.

The recycling requirements according to GS 93024 shall be complied with.

4.5 ESD drawing note

Each assembly that contains electronic components shall be marked as ESD-sensitive in accordance with GS 95009.

4.6 Amendments

Any product or process change shall be agreed with the specialist departments responsible. Such a change may be used after completed validation.

In the event of changes to existing assemblies and electrical and electronic units, the qualification scope shall be determined using the Hardware Delta Qualification Matrix (HW-DQM) and agreed with the BMW specialist department, see GS 95039 for details.

The technology validations and risk assessments required in the HW-DQM shall be carried out and verified by the supplier.

The HW-DQM represents the minimum requirement regarding qualification effort. The qualification shall be performed with electrical and electronic units produced on series production systems at the series production location and under series production conditions.

5 Production

5.1 General requirements - production

All production processes shall occur automatically. These processes shall be qualified in terms of their process capability.

The deployed employees shall be qualified for the activities to be performed. Verification shall be provided.

During the entire production, including the tests, neither components nor circuit carriers shall be electrically, thermally or mechanically overloaded, subjected to preliminary damage or destroyed (e.g. delamination, breakage, electrical, thermal or mechanical overload).

Circuit carriers that have been produced during a system malfunction shall be blocked. Further usability shall be defined and documented.

5.1.1 Cleanliness

The contamination (e.g. dust, particles) of circuit carriers between the individual production steps shall be prevented (e.g. by covering transport routes or equipment).

During the entire production process, it shall be ensured that circuit carriers and components do not come into contact with hand sweat, grease or similar. The requirements according to VDA Volume 19 T1 and VDA Volume 19 T2 shall be complied with.

Contamination (e.g. due to mounts, supports, masking) shall be ruled out.

Cardboard boxes and wooden pallets are not permissible in the SMD productions area.

NOTE Cleanliness and order are the basic requirements for smooth processes and reliable products in electronic production.

5.1.2 Climate

The climatic conditions in production (temperature, humidity, pressure in clean room areas) shall be defined and kept constant. The climatic conditions shall be documented.

Fulfillment of these specifications shall be ensured throughout the entire supply chain.

J-STD-001 may be used for orientation purposes.

5.1.3 ESD protection

A comprehensive, complete, regularly verified and documented ESD protection in the entire production process according to DIN EN 61340-5-1 and DIN EN 61340-5-2 is required.

This includes regular review and documentation of employee protection (e.g. shoes, bracelets).

The ESD check shall be carried out in conjunction with the access check of the individual person; traceability shall be ensured.

5.1.4 Traceability

Traceability for circuit carriers shall be continuously provided; see GS 95017.

5.1.5 Process change

Every process change shall be coordinated with the responsible QMT and only implemented after completed validation, see GS 95039.

5.1.6 Production relocation

5.1.6.1 Site relocation

The specifications according to GS 95039 shall be met.

5.1.6.2 Line relocation

The specifications according to GS 95039 shall be met.

5.2 Solder paste printing

5.2.1 Storage and use of solder paste

The solder paste shall be stored according to the manufacturer's specifications.

In the event of refrigerated storage, the solder paste shall be taken from the refrigerator in good time before processing. The packaging unit may not be opened until after the solder paste is completely warmed up.

After opening the packaging, it is permitted to use the solder paste only within the time specified by the solder paste manufacturer.

5.2.2 Paste application

It shall be ensured that the correct solder paste is used for each assembly. The solder paste (type, batch) shall be added to the traceability system.

Mixing or cross-contamination of different pastes (e.g. manufacturer, alloy, flux) by materials and tools used (e.g. spatula, squeegee, stencil) shall be prevented.

The solder paste is applied via stencil printing. Other application procedures (e.g. dispensing) are used only in exceptional cases. The special characteristics of the respective process shall be observed and validated with suitable measures.

Registration (alignment of the printed circuit board with the stencil) is carried out automatically with the printer's built-in camera.

Repeated solder paste printing (wet in wet) is not permissible. Exception: smooth over twice without lifting the stencil.

The paste is refilled automatically. It shall be ensured that the refill cycle is selected correctly and there is always sufficient, but never too much, paste on the stencil.

5.2.3 Printing monitoring

The print results shall be checked 100 % via SPI; see section 6.3.1.1.

If incorrectly printed circuit carriers are cleaned, it shall be ensured that the solder paste is removed without leaving behind residue. This also applies to plated through holes. Wiping is not permissible!

The cleaning process shall be approved by the QMT, see also IPC-7526.

5.2.4 Stencil cleaning

During production, the stencil reverse side undergoes automatic cyclic cleaning in the screen printer. A suitable cleaning cycle shall be defined.

The stencil cleaning after production shall be performed in an automatic stencil cleaning system.

NOTE With manual cleaning, the risk of damage and remaining residues is too high.

5.2.5 Stencil service life

The maximum permissible printing cycles of a stencil shall be defined.

A suitable procedure to determine the service life of the stencil shall be verified.
The stencil shall be sorted out after the service life has elapsed.

During ongoing production, the quality of the stencil shall be verified via regular and appropriate checks.
In case of doubt, the stencil shall be replaced early.

5.3 Assembly process

5.3.1 SMD

SMD components are fitted automatically only. The feed usually takes place in the belt or tray.

The basic requirement for a high and repeatable assembly accuracy is the precise feed and positioning of the components in the pick-up position.

When the components are mounted, the circuit carrier shall not be bent. This shall be ensured via appropriate support. No components shall be installed at the support positions and in their vicinities.

In case of assembly systems with moving X/Y table, the fitted components shall not slide from the fitting position due to the table's acceleration.

Refilling and changing of components shall be documented specifically for each assembly via the traceability system. This also applies to connecting the belts (splicing).

After conversion of the assembly line, the assembly shall be approved by a suitable check, e.g. AOI.

5.3.2 THT / Special components

Wired components are automatically fitted. Suitable automated fitting machines shall be provided for this purpose.

If, during automatic fitting of wired components the wire ends are bent (clinched), it shall be ensured that there are no short circuits due to the bent wire, even if the tool is incorrectly set.

Components are fitted manually only in exceptional cases. The mix-up and polarity reversal of components shall be prevented via appropriate measures.

5.4 Soldering

5.4.1 General requirements - soldering

All components shall be soldered according to the manufacturer's instructions.

Criteria: preheating, temperature gradient, peak temperature and time, maximum temperature, cooling speed.

The specifications of the solder paste manufacturer/ flux manufacturer shall also be observed.

Criteria: preheating, temperature gradient, soldering temperature, peak temperature and time, maximum temperature, cooling speed.

Soldering processes (rework) shall not be repeated.

MSL (moisture sensitive level) specifications shall be observed. Before each soldering process, it shall be ensured that critical components (e.g., multi-pole components, BGA, high-layer multilayers, flex-rigid or flex printed circuit boards) are dried as required according to the manufacturer's instructions.

After soldering, component connections and wire ends shall no longer be shortened.

Manual soldering is not permissible. Deviations shall be approved by the responsible QMT.

Re-soldering shall be avoided. If necessary, the process and scope of the rework shall be agreed with the QMT responsible.

5.4.2 Process monitoring

The critical process parameters shall be monitored and the measurement results documented, e.g. SPC.

The conveyor system in the soldering system shall transport the circuit carrier without jerking or slipping. Transitions shall be made steplessly.

Congestion of circuit carriers shall be prevented both in the system and at its outlet.

5.4.3 Voids in solder joints

With the completely soldered assembly, the voids in the solder joints shall not exceed the limits indicated in Table 1.

Table 1: maximum permissible number of voids

	Individual void[1)] % max.	**Void percentage total** % max.	**Comment**
BGA	20	30	Reference IPC-A-610 Evaluation using x-ray methods
Surface contact components	30	35	State of the art
Power component with high power dissipation	25	35	Agreement may need to be made based on the application.
Die attach in the power module	3 or max. 1 mm ∅	5	100 % evaluation by means of X-ray required
Purely mechanical fixation	40	50	-
1) measuring procedure: Evaluation of the projected area of X-ray analysis			

5.4.4 Reflow soldering

5.4.4.1 General requirements - reflow soldering

Reflow soldering is always carried out under an inert gas atmosphere.

Ensure that the temperature distribution on the circuit carrier/ panel is uniform.

The temperature profile shall be configured using an appropriate method of measurement.

The following framework conditions shall be taken into account:

- components with low heat capacity:
  The housing shall not exceed the maximum permissible temperature (specification by component manufacturer).
- components with high heat capacity:
  All solder joints shall reach the minimum temperature at the solder joint (specification by solder paste manufacturer).

The soldering profile shall be checked and documented in conjunction with the temperature cross profile at suitable intervals. The documentation shall be presented to the QMT if necessary.

5.4.4.2 Surface contact components

Surface contact components, e.g. BGAs, QFNs and DFNs, place significantly higher requirements for process control and monitoring. The critical process steps shall be defined and qualified.

Suitable validation options, e.g. 3D-SPI or AXI, shall be planned in the process.

5.4.4.3 Pin-in paste

The solder paste printing shall reliably provide the required solder volume.

For requirements for components, see GS 95011-1, section “General requirements – components”.

The requirements set out in section 5.4.5 shall be fulfilled.

5.4.5 Wave soldering

The flux application shall be defined in terms of quantity and application area and be carried out in a reproducible manner. The quantity shall be kept to a minimum. The flux is applied by spraying. Foam fluxing is not permissible.

If soldering frames/ masks are used, flux shall be prevented from penetrating underneath. The soldering frames or transport fingers shall be cleaned regularly.

The wave height shall be regulated and measured regularly.

Components that only rest on the printed circuit board may need to be secured against floating up. During soldering, plugs shall be secured in such a manner that they rest flat on the printed circuit board after soldering.

The solder filling at THT solder joints shall fill the plated through holes at least 75 %. The different thermal conductivity of the connection pins shall be observed (e.g. Cu and Fe connections). The THT solder filling shall be monitored as a parameter during series production.

The solder shall be checked in defined intervals in terms of its metallic constituents and the results documented. The applicable limit values (solder manufacturer) shall be complied with.

5.4.6 Selective soldering

The requirements from section 5.4.5 apply in the same manner.

All process steps shall be performed automatically.

Adjacent solder joints that have already been soldered shall not be melted again.

It shall be taken into account that the flux is generally less deactivated during selective soldering compared to wave soldering.

5.4.7 Soldering flexible circuit carriers

The special characteristics of flexible circuit carriers (bendable material, water absorption, temperature resistance) shall be taken into account via appropriate measures; see GS 95011-1, “Soldering flexible circuit carriers”.

Due to the high water absorption, PI shall be dried out before each soldering process. This also applies to flex-rigid printed circuit boards if they contain PI.

5.4.8 Soldering 3D MID

Due to the thermoplastic base materials, a soldering procedure with the lowest-possible temperature load shall be selected. Verification of process-reliable soldering shall be provided.

5.5 Cleaning/no-clean

5.5.1 General requirements - cleaning/no-clean

The state of the art is the no-clean process. Cleaning may be necessary for special requirements.

Residual contamination shall be determined by means of a suitable variable (e.g. NaCl equivalent measured in accordance with IPC-TM-650) on the qualification assemblies.

The residual contamination of the qualification assemblies shall be defined as the limit value for the series process and shall no longer be exceeded. Residual contamination shall be monitored as an SPC measure.

5.5.2 No-clean process

The compatibility of the residual contamination with the following processes/ materials (e.g. conformal coating) and their requirements shall be validated. This is done via the condensation test according to GS 95024-3-1, K-15a or by performing the climate test according to GS 95024-3-1, K-15b.

Open contacts, e.g. connector pin or contact surfaces, shall not be contaminated by flux residues, and also not by capillary effects.

5.5.3 Cleaning

If assemblies are cleaned, this shall be done immediately after the soldering process.

Complete drying of the assemblies under the components is also essential. This applies in particular to water-based cleaning and subsequent conformal coating.

Open contacts, e.g. connector pins or contact surfaces, shall not be contaminated by cleaning (e.g. transfer of contaminations).

The cleaning medium used shall be compatible with the circuit carrier and the components.

5.5.4 Brushing

5.5.4.1 Application

Brushing may only be done in the area of THT solder joints. Reflow solder joints shall not be brushed.

This process may only be used for removing solder beads after wave or selective soldering.

Assemblies in which the area of the wave and selective soldering is conformal coated are not brushed.

5.5.4.2 General requirements - brushing

Brushing is permissible only under dry conditions.

The soldering surface shall not be altered mechanically (e.g. no grooves).

The brushes shall be regularly cleaned or replaced.

Under no circumstances shall the beads that are removed fall into the brushed control unit or any other control unit.

ESD protection shall be ensured and checked regularly (e.g. ESD brushes).

5.5.4.3 Brushing procedure

Automatic: as standard (100 %, preferably overhead).

Manual: selectively, as required (e.g. after visual inspection).

5.6 Adhesion bonding

The applied adhesive volume shall be monitored.

The adhesive shall fully cross-link during curing.

The maximum permissible assembly force shall be defined and monitored in the process when joining ceramic substrates (e.g. gluing into the housing).

5.7 Bonding

5.7.1 Bond parameters

The bonding parameters (windows) for the specified material pairing, bonding method and bond loop geometry shall be evaluated and defined at an early stage – but no later than by the C-sample - by means of design of experiment (DoE) for the main influencing parameters (force, US power, bonding time, temperature if necessary).

All characteristics defined in accordance with DVS 2811 (e.g. pull and shear results, visual characteristics and, if applicable, phase occupancy tests) shall be taken into account and characterized in this process.

When bonding on active semiconductors, the parameter limits for cratering and non-stick-on-pad (NSOP) shall also be evaluated; the semiconductors shall be assessed by means of a cratering test in accordance with DVS 2811.

The limits of the determined parameter windows shall be verified and validated, and released for series production. The release documents shall be presented on request.

Deviations from the limit values according to DVS 2811 shall be proactively communicated to the customer and are subject to release by the customer.

5.7.2 Bond check

For a process-assured and reliable bonding result, the bonding parameters shall be monitored comprehensively by sensor-supported online monitoring of the welding process (see e.g. DVS 2811)

In addition, the correct execution of bond connections shall be demonstrated during series production (at least per production day or shift change, after each product change, after a change of the bond tool, wire, printed circuit board or punch grid batches, and after any adjustment of the bonding parameters within the released process window).

To do so, the following methods with the respective requirements shall be used:

Wire Bond Pull (WBP):

- the quality of wire bond joints shall be monitored during series production by means of bond pull tests in accordance with DVS 2811 and evaluated in accordance with the limit values defined therein;
- in addition to the pull forces normalized by the angle correction factors, the WBP codes and fracture patterns shall also be evaluated and documented in each case (specification of mean value, standard deviation, minimum value, maximum value; the specification limits and c_{pk} values shall be given);
- the short-term process capability index for automotive applications shall be at least c_{pk} = 1,67;

Wire Bond shear test (WBS):

- the quality of wire bond joints shall be monitored during series production by means of bond shear tests in accordance with DVS 2811 and evaluated in accordance with the limit values defined therein;
- in addition to the shear forces normalized by the angle correction factors, the shear patterns and shear codes shall also be evaluated and documented in each case (mean value, standard deviation, minimum value, maximum value; the specification limits and C_{pk} values shall be given); the nominal ball or bond root sizes for the process shall to be specified;
- the short-term process capability index for automotive applications shall be at least c_{pk} = 1,67;

Visual bond inspections:

- the quality compliant implementation of wire bonds, and the suitability of the bond surfaces, shall be visually monitored during series production in accordance with the requirements of DVS 2811;
- if 100 % testing by means of post-bond AOI is not intended, then the process capability shall be demonstrated by means of a process capability index. The long-term and short-term process capability for visual characteristics is c_{pk} = 1,33.

If specification limits are violated, an out-of-control action plan (OCAP) shall be implemented and the material bonded shall be blocked in the relevant production period until the causes have been identified and eliminated and a reassessment of the conformity of the relevant material has been made.

5.7.3 Bond covering

When bonded semiconductors and bonds are covered, special attention should be paid to the following points:

- chip and wire shall be completely covered;
- the cover and the casting shall always be free of blisters;
- the cover shall be cured according to the manufacturer's instructions.

Any form of delamination at the interfaces to wire bonds or bond surfaces or cracks of the material in the area of wire bonds are inpermissible before and after stress loading (e.g. according to GS 95024-3-1).

5.8 Conformal coating

5.8.1 Area

Open contacts (e.g. connector pins, contact surfaces) and optical components (e.g. LCD, LED, MOST) shall not be wetted with protective conformal coating.

If there is a risk that the protective conformal coating wets open contacts or optical components (e.g. spatter, overspray), these areas shall be masked during conformal coating.

5.8.2 Procedure

5.8.2.1 Dip coating

The following particularities shall be considered for the dip coating:

- the conformal coat viscosity, dipping and emerging speed shall be complied with in accordance with the conformal coating manufacturer's specifications;
- components shall be completely covered also on the bottom side;
- after removal from the coating pass, the printed circuit board shall stay in vertical or slightly tilted position to drain off the conformal coating;
- as a drainage corner, the one shall be selected where there are fewer and non-critical components;
- the contents of the immersion tank shall be checked regularly for impurities. The washing off of flux residues from the printed circuit board may make it necessary to regularly replace the conformal coating in the dip tank.

5.8.2.2 Selective conformal coating

The following criteria should be observed for selective conformal coating:

- the conformal coating area shall cover the critical circuit parts in a process-reliable manner;
- a sufficient and even coating thickness shall be ensured;
- during spray painting, shading may occur due to high components, leading to insufficiently conformal coated areas.

5.8.3 Coating thickness

The coating thickness of the conformal coating indicated in the manufacturer’s specifications.

The rule applies: as thin and even as possible.

The conformal coating shall always form a continuous film over the covered components and their connections. Special attention is paid to the edges of the pins and their housing outlet, e.g. SO, SOT housing forms.

NOTE Thick conformal coat coatings tend to crack after aging.

5.8.4 Drying

The conformal coating shall be dried according to the manufacturer’s instructions. In case of incorrect drying, there is risk that the conformal coat coating may not completely dry, but only form a skin, with the solvents trapped in the conformal coating.

The conformal coating shall be completely dried. Drying of conformal coating under components shall be evaluated due to solvent content and drying times.

5.8.5 Result

5.8.5.1 General requirements - result

The protective coat shall be free of cracks, blisters, pinholes, fish eye, voids and peeling. The conformal coat may envelop multiple component connections or components together that are close to each other if the conformal coat encloses all sides.

Conformal coating-free areas under components are permissible if they are completely enclosed by the protective conformal coat on all sides.

The conformal coating shall form an even coating in terms of its thickness and surface. Particular attention shall be paid to creating a sealed conformal coat coating at the outlet of the IC pins from the housing of the semiconductors.

Impurities or inclusion of particles are not permissible.

5.8.5.2 Surface contact components

With conventional conformal coats and conformal coat procedures, it is not possible to completely conformal coat the area under these components. It shall be ensured that no microclimate is created under these components (e.g. by open vias or conformal coating that is not completely sealed all around).

5.9 Casting/overmolding

Critical parameters:

- no solder joints shall be melted during the casting/overmolding;
- in some cases, preheating shall reduce the thermal stress on the components;
- process-reliable mixing for multi-component systems is a basic requirement;
- encasements shall always be created blister-free;
- the material shall be completely hardened or cross-linked after the injection or hardening process.

5.10 Mechanical loads

5.10.1 General requirements - mechanical stress

Assembled and soldered circuit carriers shall not be bent or subjected to a mechanical load that could damage or preliminarily damage circuit carriers, solder points or components during any handling or production step.

All production processes and handling steps shall be taken into account here.

Critical process steps

- assembly of the components,
- separation from the panel, ICT contacting,
- insertion of press-fit contacts and
- housing assembly. Particular attention shall be paid to impulse loads.

For all production processes, starting from solder paste printing on the second assembly side, reliable support of the circuit carriers or panels shall be ensured. Bending and sagging of circuit carriers or panels shall be prevented.

Mechanical load especially damages MLCCs (cracking of ceramics) and BGAs (e.g. cracks of the solder joint or damage to the package).

NOTE Both defects are typically only noticeable in the field.

5.10.2 Measurement of bending strain

If BGAs and MLCCs are arranged in critical areas, e.g. on the printed circuit board edge, near fastening points, the bending strain in this area shall be measured in three directions (0°, 45° and 90°) using strain gage measurement strips (DMS) for all production processes after the first reflow soldering.

For these measurements, the bending strain shall not exceed the limit values of the component manufacturers. If there are no separate specifications from the component manufacturers, the maximum permissible degree of bending strain is 500 µm/m.

Verification according to IPC-9704.

No process reliability is ensured with a single measurement. Verification shall be provided for process series and annual re-qualification.

Appropriate measures shall be derived if necessary.

5.10.3 Press-fit process

The press-fit procedure shall be carried out as a semi- or fully automatic. The insertion force and path parameters shall be monitored for each press-fit procedure and the specified process window or envelope curve shall be complied with.

Press-fit and support tools as well as the process shall be adapted and released for the press-fit pin.

Secure support during the press-fit process and suitable retention of the printed circuit board shall be ensured when lifting the tool. During the press-fitting, the printed circuit board shall not bend up or down.

The correct adjustment of support elements and hold-down device shall be checked in defined intervals.

All single pins shall be positioned vertically and be centered in the plated through hole of the circuit carrier.

The tumbling circle shall comply with the required tolerance.

5.10.4 Depaneling

During depaneling, torsion and impact loads shall be reliably ruled out. Particular attention shall be paid to MLCC.

Methods for depaneling and boundary conditions:

- milling is the preferred procedure. Dust extraction is required;
- roller cutters against a fixed stop are permissible;
- punching is only permissible upon agreement with the responsible specialist department and QMT;
- separation from the panel without qualified devices is not permissible, e.g. manual breaking;
- in laser cutting processes, the carbonization of the cutting areas shall be considered.

Open contacts (e.g. rubber switch mats) and visible parts (such as LEDs) shall not be contaminated with residues from the separation process (e.g. glass fibers, deposits from the laser).

5.10.5 In-circuit test (ICT)

The spring-loaded needles for contacting the ICT adapter exert pressure on the test specimen. This pressure increases with the number of needles.

Through appropriate counter bearings and holding down clamp, it shall be ensured that the test specimen is not bent during contact, e.g. breakage of MLCC, BGA solder joints.

The supports shall not be placed on components.

The adjustment of the counter bearing shall be checked after each setup and in defined intervals during the production.

5.10.6 Cutting out and bending leadframes

When cutting out overpressed or overmolded leadframes, any bending of and impact loads on the overpressed or overmolded components shall be ruled out.

This requires reliable holding-down and relief of the individual pins during the punching and bending processes.

5.10.7 Screwing

During screwing, metal chips may be produced. These shall not have any electrical, mechanical or visual impact on the electrical and electronic unit during application. Appropriate counter measures shall be provided.

The following requirements shall be fulfilled:

- use of the correct screw;
- screwing until complete contact of the head;
- preventing tilted approach;
- compliance with the screwing sequence;
- completeness of the specified screwing connections;
- prevention of loose screws in the housing;
- prevention of damage to the circuit carriers or components (e.g. use of masking);
- screwing domes shall have blind holes.

It is not permissible to rework a screwed connection.

Process monitoring of the screwing action shall at least meet the following requirements:

- recording of the screwing curve;
- torque monitoring;
- monitoring of the angle of rotation.

A category A screwing system shall be used.

5.10.8 Riveting

The riveting process shall meet the following requirements:

- monitoring of force, path and time;
- height calibration prior to work begin;
- prevention of damage to the assembly, e.g. by masking.

The following defects shall be reliably detected:

- rivet pin length outside of tolerance;
- hardness of rivet pin outside of tolerance;
- riveting pin missing.

5.10.9 Molding, clip-mounting and hot-caulking

The following requirements shall be met for molding and clip-mounting:

- the path of the press shall be monitored or a mechanical stop shall be present;
- not completely molded or clip-mounted parts shall be detected and sorted out in a process-reliable manner (process locking);
- the temperature shall be monitored during hot-caulking;
- melted material shall not contaminate components and contact areas.

5.10.10 Housing assembly

The assembly tool shall be designed such that bending or mechanical stress on the circuit carrier is not possible. The housing and assembly devices shall be reliably prevented from damaging or shearing off components.

After installation, the circuit carrier shall be fixed in the housing without tension, e.g. screw connection, mechanical guide/ fixation, different thermal expansion coefficients.

5.11 Technical cleanliness

5.11.1 General requirements - technical cleanliness

During the evaluation of requirements for technical cleanliness, the specifications and recommendations according to VDA Volume 19, T1 and the ZVEI “Guideline for technical cleanliness” shall be complied with.

Particles or impaired Minimum insulation distances, air and creeping paths shall be defined project-specifically and monitored during the safe launch phase.

For the minimum insulation distances, air and creeping paths, the requirements according to DIN EN 60664-1, DIN EN 60664-1 SUPPL1, DIN EN 60664-4 and IPC-2221, IPC-2222 and IPC-2223 shall be observed.

5.11.2 Critical particle size

The size of the solder beads or conductive particles shall not exceed 50 % of the smallest insulation distance between conductors and elements on the assembly.

5.12 Rework and repair

Rework and repair on assemblies and control units are not permissible.

Deviations from this shall be approved in scope and frequency by the QMT responsible under the following conditions:

- the type and scope of the permissible rework shall be defined in relation to the component or product. The rework process shall be defined as binding in work instructions and process descriptions;
- repeated rework of the same defect and the same circuit part is not permissible;
- the repair is generally carried out in uncoated production condition. In this process, it shall be ensured that the repair is performed in a qualified manner and there is no damage to, impairment of or preliminary damage to components, the circuit carrier or adjacent components and solder joints;
- repair after conformal coating is permissible only if the manufacturer has demonstrated a qualified repair process including conformal coating;
- rework and repairs shall be documented in relation to the product;
- trained employees are required for rework and repairs;
- the workplace shall be equipped according to the requirements, e.g. ESD protection, soldering equipment, tools, lighting, microscope;
- rework and repairs may only be carried out by employees who have been trained according to IPC- 7711/21 and have certificates that are not older than 2 years.

6 Tests

6.1 General requirements - tests

For all test steps, the check loop shall be kept as short as possible. Ideally, the tests are carried out immediately after the previous production step.

The supplier ensures that each assembly is tested with the required test depth and testing accuracy.

Skipping individual test steps shall be avoided by means of suitable measures (e.g. OK bit in the controller).

The last production step prior to packaging is the electrical function test with the concluding leakage current test of all relevant connections. The assembly is not marked or labeled until after the function test has been passed. Otherwise, an automatic check of the already existing part number shall be carried out.

6.2 Test concept

For the process approval, the following shall be available:

- test strategy, agreed upon with the QMT and development;
- design for testability analysis (DFT).
  The first DFT is performed before creating the layout and revising for the various sample phases;
- test matrix:
  This matrix serves as comprehensive test coverage. It indicates which component is tested in which test, e.g. AOI, ICT, function test, and for which parameters;
- the test coverage shall also include other function-relevant properties, e.g. thermal connection, EMC shielding;
- deviation list – this lists components that have not been tested or not sufficiently. For all components, a risk assessment shall be created from the deviation list and a process validation shall be planned.

For complex assemblies, test methods shall be selected which may check all requirements for the assembly, e.g. boundary scan.

These systems shall already be planned in the system design and circuit design.

6.3 Visual inspection

Various visual inspections are employed to ensure the required production quality and detect inconspicuous defects early on.

6.3.1 Automatic optical inspection (AOI)

The procedure is used after various production steps and in different forms.

For critical or electrically non-measurable components, the identity or polarity of the components may be detected using the “optical character reader” option.

6.3.1.1 Solder paste printing

The solder paste printing shall be 100 % checked by means of SPI (solder paste inspection).

Requirements for testing the solder paste printing:

– fine pitch, BGA and concealed solder joints: cover and coating thickness using 3D methods are required.

6.3.1.2 Assembly of components

The assembled components are tested before reflow soldering only in individual cases.

Inspection criteria are

– presence,
– position and
– polarity of the components.

6.3.1.3 Solder joints

After soldering, the circuit carriers shall be inspected using an automated process (e.g. AOI).

Criteria

– soldering quality,
– solder bridges,
– presence,
– position and
– polarity of the components.

The test limit shall be defined so precisely that test slippage is ruled out. This makes “pseudo failures” inevitable.

The detected defects shall be evaluated and classified by specially trained personnel at a subsequent test station. The aim is to keep the pseudo-failure rate as low as possible to avoid uncertainties in the subsequent assessment and test slippage.

An failure catalog with “Good - Borderline - Poor“ solder joints shall be present at the inspection workplaces.

Borderline solder joints are not reworked.

After rework or repair, a thorough visual inspection is required for

– soldering quality,
– impairment of adjacent older joints,
– solder residue and
– flux residue.

Solder residue (tin residue) on the printed circuit board is not permissible.

The results of the AOI and follow-up check shall be documented and used for continuous process optimization. The shortest-possible control loop shall be ensured.

6.3.2 X-ray inspection (AXI)

Components with concealed solder joints, e.g. BGA, DFN and QFN or exposed pad with, e.g., DPAK designpackages, should preferably be subjected to a 100% X-ray test. Only this procedure is used to test concealed solder joints.

In the case of highly integrated circuit carrier structures with SiP designs or components, a 100% in-line X-ray test may no longer provide sufficient information.

In this case, a reliable evaluation of BGA balls is no longer possible.

Here, randomly performed X-ray inspections using 3D-AXI systems are to be accompanied by regular destructive tests, e.g. metallographic sections, during the development and safe launch phase.

The random sample sizes and electrical and electronic unit assembly events shall be coordinated with QMT on a project-specific basis.

Random sample sizes are continued in reduced form in parallel to series production.

Deviations shall be validated with appropriate measures and coordinated with QMT.

6.3.3 Visual inspection

A visual inspection is not permissible.

Exceptions shall be approved by the QMT responsible.

6.4 Electrical tests

The electrical function test (preferably with original loads) shall cover the entire functional scope.
It is carried out as the last process step.

Minimum requirements for the electrical test:

- analog inputs/ outputs:
  Recording of a minimum of three measured values, at least one of which in the operating point;
- digital inputs/ outputs,
  check both logic states with worst-case thresholds;
- check bus communication;
- check the program status.

6.4.1 In-circuit test (ICT)

During contacting, damage to the conductive structures, e.g. cross-section reduction, shall be reliably ruled out.

6.4.2 Screening / run-in

Screening/ run-in is used to detect production- and component-related early failures.

All failures in the screening/ run-in shall be analyzed and the result be used for product and process optimization.

The test rate and test scope shall be agreed with the QMT responsible.

For details, see GS 95024-3-3.

6.4.3 Active trimming

In order to reduce the tolerances of a complete assembly or a sensor, they may be actively trimmed.

This is usually done by means of a laser trimming on a resistor.

Important parameters shall be observed in order to achieve accurate trimming with long-term stability:

- the trimming resistors shall not be calibrated until then and shall have a form that is conducive to trimming;
- the laser cuts shall be positioned with geometric precision;
- to achieve the required accuracy, a suitable cutting geometry shall be selected;
- laser settings, focus and power have a decisive impact on precision and long-term stability. Therefore, the laser cut shall be checked regularly under the microscope.

6.5 Mechanical tests

In order to detect deformed contact pins, it shall check the pins as the last step before packaging.

The following shall be tested:

- position (wobble circle), bending, pushed-back and protruding pins and
- hardware status, software status and delivery code.

6.6 Evaluation of test results

Storage and evaluation of the OK/not OK test results is required for traceability in the event of a failures. This procedure shall be agreed with the QMT.

6.7 Special test to verify the process-assured connection of the inner layers for mechanically drilled through-hole plating

6.7.1 General requirements - special test

Under certain conditions listed in section 6.7.2, a suitable special test (e.g. PR 334 metallization joint in PTHs test procedure) shall be performed that goes beyond the evaluation of micrographs to rule out interconnect defects (ICD). Evidence of suitability and the random sample scope of the test method used shall be submitted to the BMW specialist department upon request or agreed upon in advance. The special test is intended to demonstrate the process-assured connection of the metal layers to the metal sleeve of the plated through hole. The test results shall be presented to the releasing specialist department on request.

6.7.2 Scope of application

The test shall be performed whenever at least one of the listed conditions is met:

- a test of the inner layer connections of multilayer organic circuit carriers is explicitly required in the component requirements specification;
- when conductive polymers are used to prepare the electroplated copper metallization of plated-through holes in multilayer organic circuit carriers;
- the requirements according to IPC-A-600 section 3.1.5.1 (back etching) are not met.

6.7.3 Scope

For the series production release, the supplier shall perform the test on series production material with a statistically significant minimum scope.

7 Packaging

7.1 General requirements - packaging

The packaging for fitted circuit carriers and assemblies shall be designed such that during transport and storage no mechanical and/or electrical damage shall occur to circuit carriers or assemblies.

Devices, assemblies or components that contain critical moving parts, e.g. disc drives or hard drives, shall be transported horizontally and with as little vibration as possible. This shall be permanently ensured with every packaging unit by means of monitoring devices, e.g. tilt and shock watch.

It shall be ensured that the content of packaging units matches the label in terms of type and quantity.

Reusable packaging shall be used.

The packaging concept shall be designed to follow VDA Volume 19 T1 and the ZVEI Guidelines with regard to technical cleanliness.

7.2 ESD

It shall be ensured that no damage or preliminary damage occurs to the components caused by ESD. This requirement applies both to the production-internal packaging and transport packaging in the supply chain.

All electrical and electronic units containing electronic parts shall be packaged in an ESD-compliant manner:

- for semi-finished products, see DIN EN 61340;
- for finished products, see GS 95009.

ESD-compliant packaging shall be labelled as such.

8 Acceptance of electronic assemblies

Circuit carriers are accepted according to GS 95011-1 and IPC-A-600 Class 2.

Electronic assemblies and mechatronic units are accepted according to this Group Standard and according to GS 95011-1 and IPC-A-610 Class 3.

9 Quality assurance

9.1 Capabilities

The proofs of capability shall be approved by QMT.

9.1.1 Measuring equipment capability

For measuring systems/ measuring machines, a measuring equipment capability of $C_{mk} > 2{,}0$ shall be demonstrated.

9.1.2 Machine capability

Production machines shall have a machine capability of $C_{mk} > 2{,}0$.

9.1.3 Process capability

A product-specific process capability certificate in accordance with GS 98000 shall be provided for all production processes.

9.2 KPI (key performance indicator)

9.2.1 Intervention limits

The agreed 0-km and field control-intervention limit shall be reached by SoP.

All process steps shall be optimized before SoP.

The strategy for achieving the agreed intervention limits, e.g. yield, pseudo-error rate, number of parts produced before SoP, shall be agreed with the responsible QMT.

9.2.2 DPMO

The SMT and soldering processes shall not exceed a DPMO value of 3,4 (6σ).

The methodology for determining the DPMO value shall be agreed upon with the QMT based on IPC-9261 and IPC-7912.

9.3 Maintenance schedules

Maintenance schedules, e.g. maintenance scope, interval, tool change shall be maintained for all machines and tools.